

AstroDB Toolkit: A toolkit for creating and maintaining small, collaborative astronomical databases

Kelle L. Cruz^{1,2,3*} and David Rodriguez^{4*}

¹ Hunter College, City University of New York, United States ² Graduate Center, City University of New York, United States ³ Astrophysics, American Museum of Natural History, United States ⁴ STScI, USA
* These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) ↗
- [Repository](#) ↗
- [Archive](#) ↗

Editor: [Open Journals](#) ↗

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

We introduce the AstroDB Toolkit to fill a gap in the data management/sharing ecosystem for astronomers working on catalogs of astronomical sources. The AstroDB Toolkit is an open-source, openly developed tool that lowers the technology burden on astronomers and empowers them to make databases of astronomical sources using a common, interoperable framework. The Toolkit uses GitHub's features and its collaborative workflow.

Statement of need

In the “big data” era of large datasets and archive science, astronomers find themselves managing increasingly large and unwieldy collections of astrophysical parameters, both observed and derived. Currently, astronomers reinvent the wheel, spending time making technology choices, database design decisions, and web applications as opposed to focusing on the analysis and physical interpretation of actual data. The AstroDB Toolkit greatly lowers the technology burden on the astronomers and empowers them to make databases of astronomical sources using a common, interoperable framework.

The AstroDB Toolkit aims to serve the needs of individual astronomers and small-to-medium sized collaborations who need a data management system for hundreds to thousands of sources. The Toolkit bridges the divide where a shared spreadsheet is insufficient, but the dataset is either still living (e.g., follow-up observations are underway, new parameters being derived) or otherwise not appropriate for an institutional archive.

State of the field

The astronomical community has produced numerous data compilations, but they fall into two broad categories: large institutional archives and small grassroots efforts — and neither category offers a reusable framework for building new compilations.

Institutional archives such as the NASA Exoplanet Archive ([Akeson et al., 2013](#)), HITRAN ([Gordon et al., 2022](#)), and WISEREP ([Yaron & Gal-Yam, 2012](#)) provide excellent data access and, in the case of the Exoplanet Archive, gold-standard interoperability via the IVOA Table Access Protocol (TAP) and ADQL. However, their backends are undisclosed, they are operated by dedicated engineering teams, and they cannot be deployed by individual researchers or small collaborations. Contributing data requires submission through web forms or bulk upload APIs. These platforms are powerful for data consumers but offer no path for an astronomer to build their own analogous database.

Grassroots compilations span a wide range of technology choices with little interoperability between them. The Open Astronomy Catalogs (AstroCats; Guillochon et al. ([2017](#))) represent

the most mature open-source effort: each transient source is stored as a richly annotated JSON file in a Git repository, with a documented schema that mandates source attribution for every data point. The framework was explicitly designed to be reusable, and four separate catalogs (supernovae, tidal disruption events, novae, black holes) were built on it. However, AstroCats supports only JSON — there is no relational query layer — and the codebase targets Python 2/3.4–3.6, limiting its continued maintainability. The MOCA database of open clusters (Malo & Gagné, 2021) uses a hosted MySQL backend with a companion Python package (mocapy) for querying, but the database schema is not designed to be forked and repopulated for a new science case. The Community Atlas of Tidal Streams (Price-Whelan et al., 2021) encapsulates its science logic in Jupyter notebooks with no persistent database backend, making it difficult to query programmatically or extend to new data types. At the lightest end of the spectrum, compilations such as the Ultracool Sheet (Best et al., 2025) and the Hypatia Catalog (Hinkel et al., 2014) rely on Google Sheets or web-only interfaces — convenient for human browsing but lacking version control, contribution workflows, or machine-readable schemas.

Across this landscape, several gaps are consistent. Schema documentation is rare. Contribution workflows are ad hoc (CSV uploads, pull requests against JSON files, or web forms), with no shared tooling. Most critically, none of these projects offer a general-purpose, deployable framework: an astronomer who wants to build their own compilation of, say, low-mass stars or high-redshift galaxies cannot fork any of these projects and adapt the infrastructure to their science case without essentially starting from scratch.

Software design

The technical choices that were made in the development of the AstroDB Toolkit followed several design requirements:

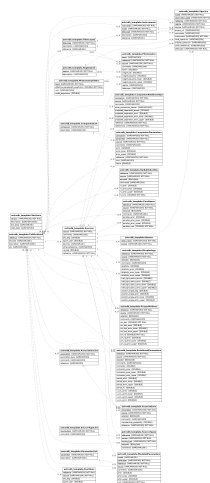
- Ability to represent complex data models in a simple, intuitive fashion,
- Support for datasets up to tens of thousands of astrophysical sources,
- Collaborative editing that enables community members to modify and maintain data holdings,
- Be usable privately for managing not-yet-public data,
- Interactively explore and visualize holdings via a website,
- Queryable locally via scripting without the need for an internet connection.

As described below, the AstroDB Toolkit meets all of these requirements.

The AstroDB Toolkit consists of:

- a template schema ([astrodb-template-db](#)),
- Python packages ([AstrodbKit](#), [astrodb_utils](#)), and
- [documentation and suggested workflows](#)

The AstroDB Toolkit facilitates the creation of intermediate-scale databases focused on typical astronomy use-cases. The Toolkit's template schema (??) relies on an object-model that naturally translates to astronomical sources, that is to say, the core table in the database can represent objects such as brown dwarfs or galaxies, while supporting tables represent properties of that object. Because the AstroDB Toolkit's core Python package, *AstrodbKit*, is a wrapper around the database package *SQLAlchemy*, it supports a large range of database architectures, including *SQLite*, *PostgreSQL*, *MySQL*, etc, while using language that is familiar to astronomers. *AstrodbKit* also supports cone searches and can output results as *astropy.table* objects.



84 Uses Felis.

85 Collaborative Workflow and Testing with GitHub

86 One of the key design requirements for an AstroDB Toolkit-powered database is support for
87 collaborative editing of the holdings using a GitHub workflow. As a result, the Toolkit creates
88 databases that are fundamentally a set of plain text JSON files that describe each object.
89 When users make changes to the properties of an object, such as adding a new spectrum
90 or updating a value for a radial velocity, these changes are human-readable as a simple diff
91 between two JSON files and can be reviewed via pull requests. This JSON document store
92 architecture allows for a community to maintain a database, review changes as they come
93 in, and use automated tools to validate the database. The AstrodbKit package has tools
94 to readily transform data between the document store and relational database to facilitate
95 managing local, private data as well as external applications, such as a hosted website.

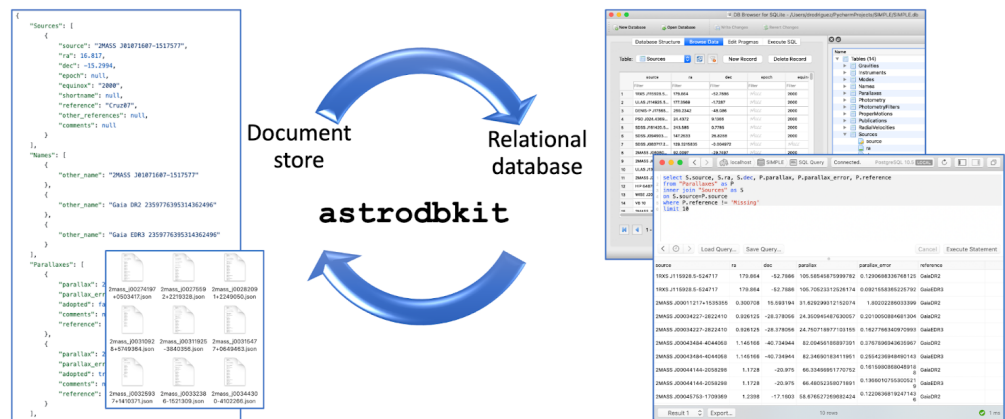
96 An individual user may deploy their own copy of any AstroDB database. They may make
97 changes in their local branch and push to their copy on GitHub. By issuing a pull request, they
98 request their changes be adopted into the main branch of the database. Because the database
99 is stored as individual JSON documents, reviewers can see exactly which objects have been
100 updated and can comment on the changes if needed.

101 As part of the pull request process, automatic tests implemented via GitHub Actions can be
102 run to verify the integrity of the database. This ensures no changes took place that break the
103 functionality of the database and also include verification for the data that has been added.

104 Finally, when the pull request is accepted, additional automated tasks can be performed to
105 regenerate the database and push it to external users of the database, such as a hosted website.

Document store of
plain text JSON files

Relational database



Examples of handling SIMPLE as a document store, a collection of ASCII JSON files.

Examples of handling SIMPLE as a relational database, using DB Browser for SQLite and

Figure 1: Toolkit design.

106 **Spectra and non-tabular data**

107 Spectra, images, and other non-tabular data are stored as pointers to cloud-hosted files. The
108 files are hosted in a variety of places, including institutional repositories. However, we have
109 found the easiest and most reliable host to be Amazon Simple Storage Service (Amazon S3).
110 Amazon S3 provides individual links to each file and since they are infrequently accessed, the
111 cost is minimal. In order to enable the database to fully function without an internet connection
112 and/or to point to files not hosted in the cloud, we allow pointers to local files using an
113 environment variable to indicate a local path. AstrodbKit contains logic to understand which
114 columns contain pointers to spectra and translates them into specutils.Spectrum objects.

115 **Research impact statement**

116 *Evidence of realized impact (publications, external use, integrations) or credible near-term*
117 *significance (benchmarks, reproducible materials, community-readiness signals). The evidence*
118 *should be compelling and specific, not aspirational.*

119 The current poster-child of the AstroDB Toolkit is SIMPLE, the Substellar and IMaged PLANet
120 Explorer Archive of Complex Objects <https://simple-bd-archive.org/>.

121 The newest usage of the Toolkit is focused on Dwarf Galaxies. There are tens if not hundreds
122 of galaxies in the Local Group, with a diverse range of applicable science problems. Many of
123 these scientific applications depend on having as complete as possible a list of properties for
124 these galaxies including total luminosity, half-light radius, and redshift/systemic velocity.

125 **AI usage disclosure**

126 Generative AI tools were used at several stages of this work.

127 **GitHub Copilot** was used to assist in writing small portions of code and inline documentation
128 throughout the development of the AstroDB Toolkit. Its assistance was limited to code
129 completion and documentation suggestions.

130 **Claude Sonnet 4.6** (Anthropic, accessed via Claude Code CLI) was used to assist in drafting
131 the “State of the field” section of this manuscript, including researching and summarizing the

132 technology stacks and schemas of related astronomical data compilations, and constructing
133 initial BibTeX bibliography entries for cited works.

134 All AI-generated text and code was thoroughly reviewed, revised, and validated by the human
135 authors. The authors made all primary architectural and design decisions for the AstroDB
136 Toolkit. The authors take full responsibility for the accuracy of all content in this manuscript,
137 including citations and bibliographic details.

138 Acknowledgements

139 We acknowledge contributions from Arjun, HS students, etc.

140 References

- 141 Akeson, R. L., Chen, X., Ciardi, D., Crane, M., Good, J., Harbut, M., Jackson, E., Kane, S.
142 R., Laity, A. C., Leifer, S., Lynn, M., McElroy, D. L., Papin, M., Plavchan, P., Ramírez, S.
143 V., Rey, R., Braun, K. von, Wittman, M., Abajian, M., ... Zhang, S. (2013). The NASA
144 exoplanet archive: Data and tools for exoplanet research. *Publications of the Astronomical*
145 *Society of the Pacific*, 125(930), 989–999. <https://doi.org/10.1086/672273>
- 146 Best, W. M. J., Dupuy, T. J., Liu, M. C., Sanghi, A., Siverd, R. J., & Zhang, Z. (2025).
147 *The UltracoolSheet: Photometry, astrometry, spectroscopy, and multiplicity for 4000+*
148 *ultracool dwarfs and imaged exoplanets* (Version 2.1.0) [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.15802304>
- 150 Gordon, I. E., Rothman, L. S., Hargreaves, R. J., Hashemi, R., Karlovets, E. V., Skinner,
151 F. M., Conway, E. K., Hill, C., Kochanov, R. V., Tan, Y., Wcisło, P., Finenko, A.
152 A., Nelson, K., Bernath, P. F., Birk, M., Boudon, V., Campargue, A., Chance, K. V.,
153 Coustenis, A., ... Yurchenko, S. N. (2022). The HITRAN2020 molecular spectroscopic
154 database. *Journal of Quantitative Spectroscopy and Radiative Transfer*, 277, 107949.
155 <https://doi.org/10.1016/j.jqsrt.2021.107949>
- 156 Guillochon, J., Parrent, J., Kelley, L. Z., & Margutti, R. (2017). An open catalog for
157 supernova data. *The Astrophysical Journal*, 835(1), 64. [https://doi.org/10.3847/1538-](https://doi.org/10.3847/1538-4357/835/1/64)
158 [4357/835/1/64](https://doi.org/10.3847/1538-4357/835/1/64)
- 159 Hinkel, N. R., Timmes, F. X., Young, P. A., Pagano, M. D., & Turnbull, M. C. (2014). Stellar
160 abundances in the solar neighborhood: The Hypatia catalog. *The Astronomical Journal*,
161 148(3), 54. <https://doi.org/10.1088/0004-6256/148/3/54>
- 162 Malo, L., & Gagné, J. (2021). MOCA: The montreal open clusters and associations catalog.
163 *Research Notes of the American Astronomical Society*, 5(9), 221. [https://doi.org/10.](https://doi.org/10.3847/2515-5172/ac2440)
164 [3847/2515-5172/ac2440](https://doi.org/10.3847/2515-5172/ac2440)
- 165 Price-Whelan, A. M., Bonaca, A., Caldwell, N., Cerny, W., Conroy, C., Cunningham, E. C.,
166 Hernquist, L., Koposov, S. E., Naidu, R. P., Nidever, D. L., Oh, S., Speagle, J. S., & Vivas,
167 A. K. (2021). Off the beaten track: A new catalog of stellar streams in the milky way. *The*
168 *Astrophysical Journal*, 910(1), 17. <https://doi.org/10.3847/1538-4357/abcb8d>
- 169 Yaron, O., & Gal-Yam, A. (2012). WISEREP — an interactive supernova data repository.
170 *Publications of the Astronomical Society of the Pacific*, 124(917), 668–681. [https://doi.](https://doi.org/10.1086/666656)
171 [org/10.1086/666656](https://doi.org/10.1086/666656)